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CCUS in Coal-Fired Power Plants: Process Innovation and Practice of CO₂ Capture and Catalytic Methanol Synthesis

Luhan Yang, Na Zhao, Yanan Huang, Jihai Yang, and Hengwei Liu, Wison Engineering Ltd., Shanghai, China

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With the accelerated advancement of global climate governance, China—currently the world's largest emitter of carbon dioxide—faces unprecedented decarbonization pressure^(Zhenyu Zhao, 2013). In 2022, energy-related carbon emissions surged to 11 billion tonnes, constituting over 80% of the national total. Emissions from coal-fired power plants alone exceeded 4.5 billion tonnes, accounting for 43% of energy-sector emissions^(Annual-Report-Working-Group-on-China's-Synergistic-Pathways-for-Carbon-Neutrality-and-Clean-Air, 2022). This structural disparity has grown increasingly prominent under the implementation of China's "Dual Carbon" strategy. China's *14th Five-Year Plan for the Modern Energy System* explicitly mandates a 20% share of non-fossil fuel consumption by 2025^{(National Development and Reform Commission (NDRC), 2022)}. In parallel, a "Three-pronged Retrofitting" initiative for coal-fired power plants is being implemented, assigning the power sector responsibility for 40% of national emission reductions—placing it at the forefront of the carbon neutrality transition. The *Action Plan for the Low-Carbon Transformation of Coal Power (2024–2027)* identifies biomass co-firing, green ammonia blending, and end-of-pipe carbon capture, utilization, and storage (CCUS) as the three primary pathways for coal power decarbonization, and explicitly includes CO₂ hydrogenation to methanol as a recognized technical route for coal-fired power retrofitting. Concurrently, the European Union's amendment to the *Renewable Energy Directive II (RED II)* recognizes methanol derived from non-renewable CO₂ captured in industrial and power generation sectors as "green methanol"—provided it is accounted for under the EU Emissions Trading System^(EU, 2018). This regulatory expansion substantially enlarges the potential market for green methanol and stimulates adoption of CO₂ hydrogenation technology across high-emission industries.

Within the policy and technological context, CO₂ hydrogenation to methanol has emerged as an innovative process pathway for carbon resource utilization, drawing considerable attention. Despite the successful deployment of several pilot projects—including Mitsui's 100-ton/year pilot plant in Japan, multiple 100,000-metric-ton-scale plants by Iceland's Carbon Recycling international^(H. Wang et al., 2020), and Geely Group's 30,000-tonne/year plant using Clariant catalysts^(NenPower, 2025)—key technical and economic barriers persist. These include limited CO₂ capture efficiency, suboptimal methanol selectivity and conversion, and inadequate system energy utilization efficiency^(Roy et al., 2025). While the coupling of green hydrogen with CO₂ by hydrogenation process is recognized as a critical pathway toward sustainable

methanol production, large-scale deployment remains constrained due to techno-economic immaturity^(Zhang et al., 2025).

This study reports on the successful engineering design, procurement, construction and testing& pilot test operation (EPC+TO) of a 15,000-ton-scale demonstration plant for CO₂ capture from flue gas and 10,000 t/a of subsequent methanol synthesis at Shanghai Waigaoqiao No.3 Power Plant by Wison Engineering. The plant integrates an AEEA-based amine absorbent for CO₂ capture with low composition and a high-performance, modified copper-based catalyst for methanol synthesis. The project addresses multiple challenges encountered in industrial-scale production of methanol via CO₂ capture & purification, offering engineered solutions through its innovative design. Employing an AEEA-based composite absorbent to enhance CO₂ capture efficiency, as well as reducing the capture consumption, the system integrates a high-activity modified copper-based catalyst for efficient methanol synthesis. Building upon the practical insights gained from this industrial implementation, the paper further explores the potential application scenarios when coupled with green hydrogen, providing valuable real references for low-carbon transformation in traditional energy sectors.

Introduction

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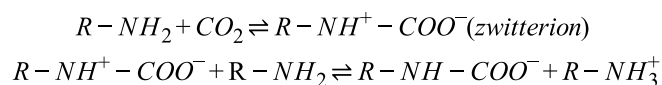
Process Development of CO₂ Hydrogenation-to-Methanol at Waigaoqiao Power Plant

CO₂ Capture from Low-Concentration Flue Gas via Amine Absorption

CO₂ capture technologies encompass absorption, adsorption, membrane separation, and cryogenic distillation^(Chai, Ngu, & How, 2022). Among the diverse carbon capture pathways, chemical absorption stands out as the predominant approach for treating flue gas with low CO₂ concentrations—such as those from coal-fired power plants and industrial kilns—owing to its high maturity and strong adaptability^(Zeng, Qu, Xu, & Luo, 2022). This method leverages selective chemical reactions between alkaline absorbents and acidic CO₂ to achieve carbon capture, making it particularly suitable for handling large-scale, low-concentration flue gas systems^(Liang et al., 2015; Xiaoxing Wang & Song, 2020). However, chemical absorption systems also face several technical barriers: degradation of solvents induced by SO_x, NO_x, and O₂ in the flue gas within the operating temperature range; high energy demand for regeneration significantly elevates operational costs^(Du et al., 2024). Consequently, developing efficient and low-energy-consuming absorbents remains the central research focus for CO₂ capture processes targeting low-concentration flue gas.

Amine-based solvents are the most extensively used absorbents for CO₂ capture^(Loachamin et al., 2024) with low flue gas concentration for high CO₂ concentration applications. Conventional industrial practices employ single-component amines (e.g., MEA, MDEA) or their mixtures^(Rochelle, 2009). While these solvents exhibit rapid absorption kinetics^(Liang et al., 2015) and relatively high CO₂ uptake capacity^(Ochedi, Yu, Yu, Liu, & Hussain, 2021), they also suffer from notable drawbacks, including strong corrosivity^(Stergioudi et al., 2022), poor thermal stability^(Khan, Halder, & Saha, 2015), and elevated regeneration energy consumption^(M. Wang, Joel, Ramshaw, Eimer, & Musa, 2015). This study employs an optimized AEEA (C₄H₁₂N₂O)-based mixed amine solution that demonstrates superior CO₂ capture performance compared to conventional MEA-based solvents in both laboratory testing and industrial applications.

The CO₂ absorption mechanism by AEEA fundamentally involves reversible nucleophilic addition reactions between amino groups and CO₂, proceeding as follows:



This reaction pathway suppresses the formation of stable carbamates typical in primary amines^(Zhou et al., 2024), thereby enhancing desorption efficiency and reducing the overall energy demand of the process. The unique trifunctional structure of AEEA molecules—incorporating primary amine, secondary amine, and hydroxyl groups—confers distinct advantages over MEA in CO₂ capture performance^(Aso, Orimoto, Higashino, Taniguchi, & Aoki, 2022): its uptake capacity reaches approximately 1.74 times that of MEA at equivalent concentration^(Sholeh Ma'mun, 2007); CO₂ absorption rates are markedly accelerated in biphasic systems containing AEEA^(F. Liu, Rochelle, Wang, Chen, & Fang, 2021); and stable kinetics persist even under high loading conditions^{(Huang et}

al., 2024). Moreover, literature indicates that the activation energy for CO₂ desorption in AEEA–CO₂–H₂O systems (32 kJ/mol) is 27% lower than that of MEA-based systems (44 kJ/mol), resulting in a 10% reduction in regeneration energy demand (Krótki et al., 2023).

Given these advantages, the AEEA-based amine solvent was selected for the CO₂ capture unit in this project. The feedstock gas utilized in this project is directly diverted from the flue duct of a coal-fired power plant into the CO₂ capture processing unit, characterized by a CO₂ concentration ranging from 18%wt to 20%wt. This constitutes a low-concentration CO₂ stream, which presents substantial challenges for effective carbon capture and consequently entails elevated energy consumption throughout the capture process. Operational results demonstrate that when absorption temperature, desorption temperature, and system pressure drop meet design specifications, the unit achieves an average capture rate of 95.1%, organic amine regeneration efficiency of 98.01%, dry-basis CO₂ purity exceeding 99%, and specific regeneration energy consumption below 2.0 GJ/tCO₂—representing industry-leading performance metrics.

Hydrogenation Catalyst

The development of high-performance catalysts constitutes a core challenge for methanol synthesis. Copper-based catalysts (e.g., Cu/ZnO/Al₂O₃) have emerged as the industrial mainstream choice due to their exceptional low-temperature (200–300 °C) catalytic activity and cost effectiveness (X. Liu, Wang, & Lu, 2024). Nevertheless, conventional coprecipitation-prepared catalysts exhibit an inherent limitation in insufficient Cu–ZnO interfacial contact, resulting in constrained active site density that impedes further improvement of catalytic efficiency (Saedy et al., 2022).

Improving the Cu–ZnO interface has been identified as a critical pathway for enhancing catalyst efficacy. In this project, a novel Cu/ZnO/Al₂O₃ catalyst developed by a research group from East China University of Science and Technology was employed (Xuguang Wang et al., 2025). Through systematic exploration of the optimal Cu–Zn interface contact and characterization of active sites, the preparation process was optimized to promote the migration of active centers toward the catalyst surface. This modification significantly enhanced catalyst activity, stability, and selectivity for methanol formation under industrial operating conditions.

Methanol Synthesis Reaction and Process Optimization

The hydrogenation of CO₂ to methanol is an exothermic reaction ($\Delta H_{298K} = -49.5 \text{ kJ} \cdot \text{mol}^{-1}$), with an optimal operating temperature range of 200–280 °C (Guil-López et al., 2019). Within this range, increasing temperature enhances CO₂ conversion and methanol yield, but typically reduces methanol selectivity. The typical operating pressure lies between 3.0 and 6.0 MPa, with higher pressures favoring forward reaction kinetics but also increasing equipment costs and material requirements. To balance catalyst activity, operational stability, and economic feasibility, this process adopts optimized conditions of 4.0–5.0 MPa and 220–260 °C. An adiabatic-isothermal shell-and-tube composite reactor is employed. A pre-heated adiabatic catalyst bed is positioned above the upper tube sheet to initiate the reaction, after which the reactant mixture enters the tube bundle for further conversion. Reaction heat is effectively removed via steam generation on the shell side. The reactor internals were optimized to maintain precise thermal control, with radial bed temperature variation maintained below 2 °C and axial variation below 5 °C, maximizing catalyst performance and ensuring long-term operational stability.

A two-stage reactor configuration was adopted to significantly reduce the recycle gas volume, thereby lowering circulation energy consumption and downsizing equipment in the hydrogenation loop—facilitating scalability for large-scale deployment. Flash steam from crude methanol separation is recycled to the CO₂ compressor inlet, reducing vent gas losses and improving feedstock utilization. Additionally, steam generated from the exothermic reaction is utilized for methanol distillation and rich amine regeneration, lowering overall process steam consumption. Since methanol distillation is a well-established industrial operation, a distillation section was not included in this demonstration project.

During the performance evaluation phase, under operating conditions of 4.0–5.0 MPa and 220–260 °C, the single-pass conversion of CO₂ exceeded 50%, methanol selectivity reached over 99%, and the overall conversions of H₂ and CO₂ after recycling were both above 99%.

Process Description

Process Overview

The process configuration adopted in this project consists of two primary units: a CO₂ capture unit and a methanol synthesis unit. In the CO₂ capture unit, flue gas from the power plant first passes through a pre-treatment tower to remove SO_x, NO_x, and other impurities. The treated gas then enters the absorption column, where it contacts the amine-based solvent in a counter-current flow, enabling efficient CO₂ absorption. The resulting CO₂-rich solvent is fed into the desorption column for regeneration, where CO₂ is released and subsequently compressed before being introduced into the methanol synthesis unit. In the synthesis section, CO₂ is mixed with hydrogen at the stoichiometric ratio and pressurized to 5.0 MPaG. The gas mixture then passes sequentially through two fixed-bed isothermal reactors arranged in series. Each reactor outlet is equipped with a cooling and separation system. The resulting crude methanol products are combined and routed out of the unit boundary, while unreacted gases are recycled to the inlet of the first-stage reactor for further conversion.

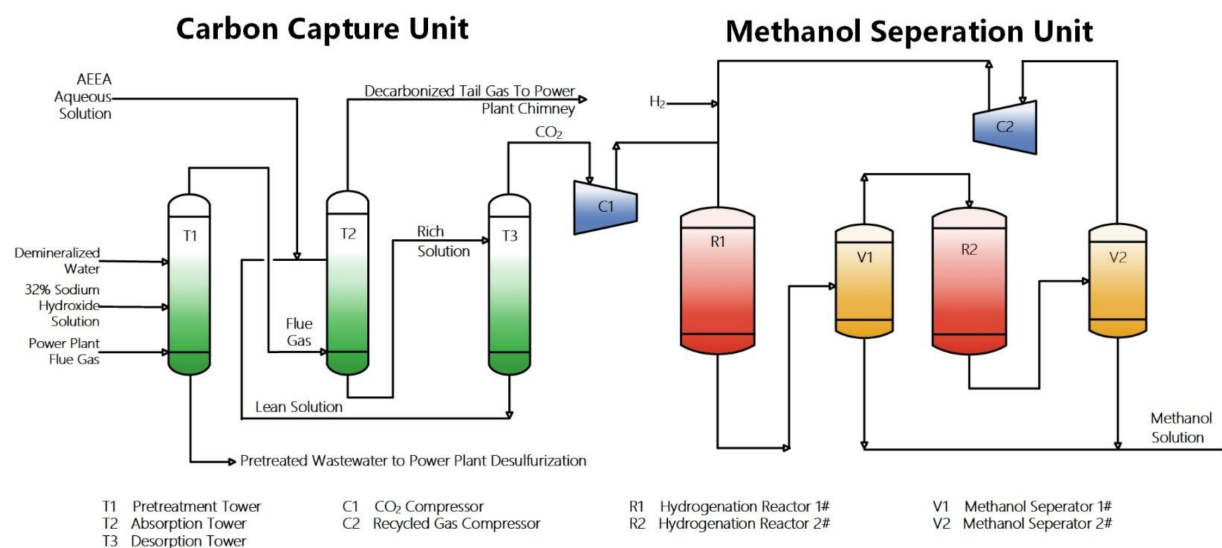


Figure 3.1—Simplified process flow diagram of the demonstration plant

Performance Evaluation

A continuous performance assessment was conducted in January 2025, during which all major process parameters—such as amine circulation rate, absorption/desorption temperature, system pressure drop in the CO₂ capture unit, as well as reaction temperature, pressure, recycle ratio, purge volume, and pressure drop in the methanol synthesis unit—remained within design specifications. Key performance metrics are summarized as follows:

1. CO₂ capture rate: Ranged from 85.22% to 99.93%, with an average of 95.10%;
2. Conversion and Selectivity: Overall CO₂ conversion reached 99.98%, H₂ conversion was 99.5%, and methanol selectivity exceeded 99.99%, with only trace amounts of methane and CO detected;
3. Production capacity: the plant operated at its full design capacity of 10,000 tonnes/year, confirming the feasibility of large-scale industrial implementation.



Figure 3.2—Site photograph of the 10,000 t/a demonstration plant at Waigaoqiao No. 3 Power Station.

Outlook on Green Hydrogen Integration and Development Pathways

Techno-Economic Analysis of Green Hydrogen-Coupled Methanol Production

In the context of climate change mitigation and carbon neutrality, coupling industrially captured CO₂ with green hydrogen produced via renewable-powered water electrolysis is emerging as a promising pathway for carbon recycling and energy transition^(Ouda et al., 2019). This integrated approach not only significantly reduces industrial carbon emissions but also yields clean products—such as fuel, chemical feedstock, or marine fuel—delivering both environmental and economic benefits^(Cao et al., 2025).

Currently, the economic viability of green hydrogen-coupled CO₂-to-methanol processes remains the primary barrier to its large-scale commercialization, with production costs still substantially higher than those of fossil-based methanol (e.g., coal or natural gas-derived). Under the current market conditions in China, this study provides a preliminary cost estimation for traditional coal-to-methanol and green hydrogen-coupled CO₂ hydrogenation processes. The coal-to-methanol price was estimated at 333.5 USD/t based on the following parameters: the coal price was taken as the average market price in July 2025, with an anthracite consumption rate of 2.0 t/t methanol; consumption of fuel coal, cooling water, and electricity was derived from literature data^(Guoqing et al., 2025); and processing plus depreciation costs were estimated at 20% of raw coal cost. As presented in Table 4.1, which incorporates the utility consumption data of this project and the projected 2025 green hydrogen production costs in China, the total cost of green hydrogen-coupled CO₂-to-methanol was estimated to be 724.4 USD/t, with green hydrogen production accounting for 81% of the total cost, representing the determining factor for the process economics.

Table 4.1—Material consumption and cost estimation per tonne of methanol in the green hydrogen-coupled hydrogenation process.

Item	Consumption	Unit Cost	Cost (USD/t MeOH)
Green Hydrogen	0.195 t/t	3.0 USD/kg*	585
Electricity	412 kWh/t	0.07 USD/kWh	28.8
Low-pressure Steam	3.395 t/t	16.74 USD/t	56.8
Softened Water	1.448 t/t	0.98 USD/t	1.4
Instrument Air	60 Nm ³ /t	0.02 USD/Nm ³	1.2
Nitrogen	12 Nm ³ /t	0.1 USD/Nm ³	1.2
Labor& Manufacturing	/	/	50
Total	/	/	724.4

*

Based on the green hydrogen price in Inner Mogolia, China

Sensitivity Analysis and Cost Competitiveness

In green methanol production, green hydrogen constitutes approximately 60%–80% of the total production cost. Although currently expensive, electrolytic hydrogen production costs are projected to decline significantly to 1.3–4 USD/kg by 2030, driven by decreasing renewable energy costs and electrolyzer technology advancements. Cost parity between green and gray hydrogen may be achieved by 2040 in emerging markets like China and India, facilitating large-scale green methanol commercialization. In any case, the cost of green hydrogen is dramatically determined by the equivalent hours of renewables (e.g. wind and solar or their consortium). This anticipated cost reduction is expected to create favorable conditions for the large-scale commercialization of green methanol. Representative green hydrogen production costs across selected countries and regions are summarized in [Table 4.2](#).

Table 4.2—Estimated Material Consumption and Cost Breakdown for Green Hydrogen-Coupled Methanol Production (USD/kg)

Region	2025	2030	2035	2040	2045	2050
Europe	4.0–6.3	3.0–4.0	2.5–3.5	2.0–3.0	1.8–2.5	<2.0
USA	3.5–6.0	2.5–3.7	2.0–3.0	1.7–2.5	1.5–2.2	1.0–1.5
China	3.0–6.0	2.0–3.0	1.5–2.5	1.2–2.0	≤1.5	<1.2
MENA	3.0–4.0	2.0–3.0	1.5–2.5	≤2.0	≤1.5	≤1.2
Latin America	3.0–6.0	1.3–2.5	1.2–2.0	≤2.0	≤1.5	≤1.2
Australia	3.0–6.0	1.5–2.5	1.2–2.0	1.0–1.8	≤1.5	1.0–1.5

With the deepening transition towards a low-carbon global energy structure, the cost competitiveness of CO₂ hydrogenation to methanol is expected to improve progressively, driven by the combined effects of declining green hydrogen prices, carbon credit revenues, and increasingly stringent carbon taxation. As schematically illustrated in [Figure 4.1](#), the techno-economic competitiveness of this pathway can be estimated using the following relationship: Competitiveness = (Conventional methanol production cost + Carbon tax) – (CCU baseline cost – Carbon credit income – Green hydrogen subsidies). Given the nascent stage of carbon offset schemes in most countries, this study focuses on a quantitative analysis of the impacts of green hydrogen pricing (subsidised values, as presented in [Table 4.2](#)) and carbon taxation on methanol production costs.

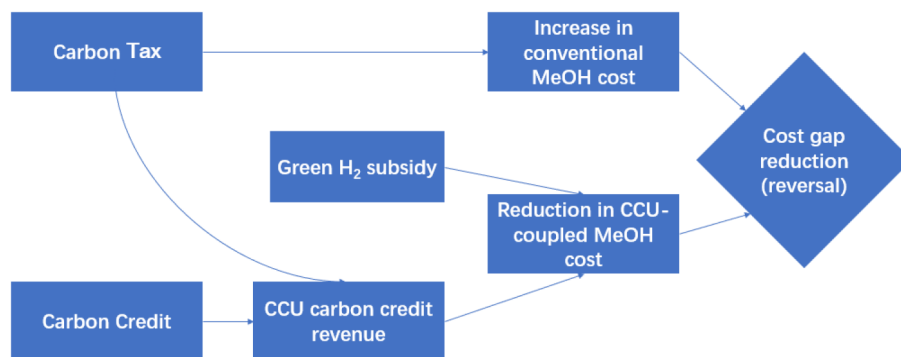


Figure 4.1—Influence pathway diagram of cost competitiveness in CO₂-to-methanol conversion via hydrogenation

In 2024, the domestic market price for premium-grade methanol in China ranged between 306 - 356 USD per tonne, with an average value of 326 USD/t adopted as the benchmark cost for coal-derived methanol in the 2025 baseline scenario illustrated in Figure 4.2. As shown in Figure 4.2a, under a scenario where carbon taxation increases by 50% every five years, the production cost of CO₂-derived methanol gradually converges with that of conventional coal-based methanol after 2045. Figure 4.2b presents the impact of declining green hydrogen prices under constant carbon taxation. In this case, the production cost of CO₂-derived methanol decreases in tandem with falling hydrogen costs. Assuming stable coal prices, cost parity between the two production routes is projected to occur after 2035. Figure 4.2c illustrates a combined scenario in which both green hydrogen prices decrease and carbon tax rates increase over time. Under such condition, the cost of coal-based methanol steadily rises and surpasses that of CO₂-based methanol prior to 2035. A comparative analysis of Figures 4.2a and 4.2c indicates that the sustained decline in green hydrogen prices substantially accelerates cost convergence between the two pathways and serves as a key enabler in improving the economic viability of CO₂ hydrogenation. Taken together, our analysis suggests that when green hydrogen prices fall within the range of 1.2–2.0 USD/kg and carbon tax levels reach approximately 30 USD/t, the CO₂-to-methanol route will exhibit a clear economic advantage over conventional coal-based methanol production.

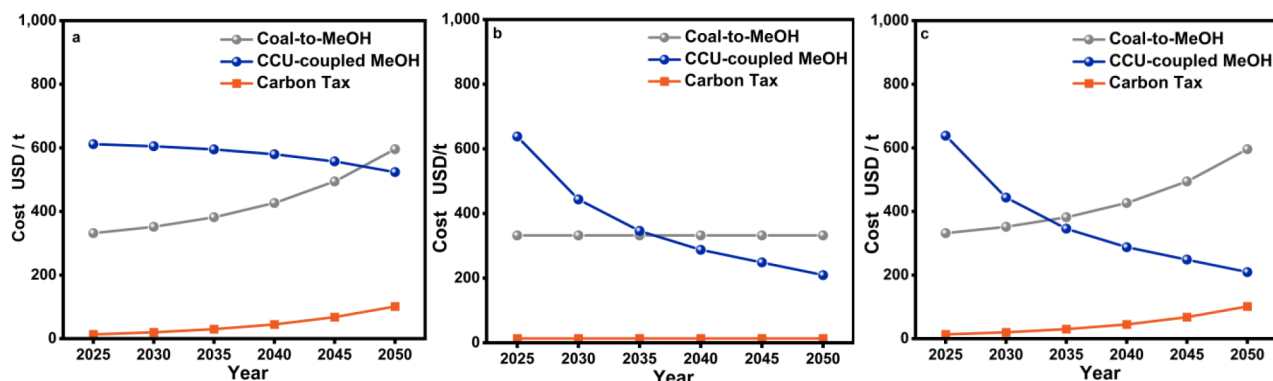


Figure 4.2—Estimated Methanol production cost. (a) Impact of carbon tax on methanol production cost; (b) Impact of green hydrogen price on methanol production cost; (c) Combined impact of green hydrogen price and carbon tax on methanol production cost.

Carbon Footprint Accounting

Following comprehensive accounting, the integrated process of green hydrogen-coupled CO₂ hydrogenation to methanol has the potential to achieve an annual CO₂ emission reduction exceeding 2.1 million tonnes for a single 600 MW coal-fired unit. This system consumes approximately 290,000 tonnes of hydrogen per year, requiring 16.2 billion kWh of renewable electricity for water electrolysis (considering the online ratio

of 55% for water electrolysis, the number of installed capacity for renewables would almost be doubled), while simultaneously producing 1.5 million tonnes of green methanol annually.

The process exhibits a cradle-to-gate carbon footprint of 300 kgCO₂e per tonne of refined methanol (equivalent to 15.0gCO₂/MJ), complying with the European Union's Renewable Energy Directive (RED) emission limit of 560 kgCO₂ per tonne of green methanol production. It is worth noting that the emission factor for electricity used in this assessment is based on the coal-fired power value from the Ecoinvent database, which is significantly higher than that of renewable electricity. Should the system be powered entirely by green electricity in the future, the total carbon footprint could be substantially reduced.

In summary, the integration of flue gas CO₂ hydrogenation with green hydrogen represents a promising and strategically important technology for achieving carbon neutrality. While current economic and technical barriers persist, ongoing advancements in core technologies and policy framework optimization are anticipated to enable broader implementation of green methanol across energy and chemical sectors.

Table 4.3—Carbon footprint assessment of green hydrogen-coupled CO₂ hydrogenation to methanol

Item	Consumption per tonne of methanol	Carbon footprint (kgCO ₂ e)	Emission factor
Captured CO ₂	1.4 t	−1400	−1.0 kgCO ₂ e/kg
Green hydrogen	0.195 t	0	0
Low-pressure steam	3395 kg	1208	0.3557 kgCO ₂ e/kg
Alkali solution	8.624 t	11	1.2904 kgCO ₂ e/t
Off-gas	/	56	/
Electricity	412 kWh	425	1.0304 kgCO ₂ e/kWh
Total	/	300	/

Scalable Production Solutions

The ongoing decarbonization and energy-efficiency retrofitting of traditional high-emission industries have not only broadened the application prospects of integrated green hydrogen-coupled CO₂ hydrogenation to methanol technologies, but also imposed stringent requirements on the cost-effectiveness and scalability of green hydrogen production and supply. Taking a single 600 MW power generation unit as an application scenario, implementation of this integrated pathway enables an annual CO₂ emission reduction exceeding 2.1 million tonnes and the co-production of 1.5 million tonnes of green methanol. Concurrently, its green hydrogen demand reaches approximately 290,000 tonnes/year, necessitating an annual consumption of renewable electricity up to 16.2 billion kWh.

Large-scale green hydrogen production necessitates urgent solutions to address core challenges such as optimization of the levelized cost of hydrogen (LCOH) and accelerated project delivery. In response, Wison Engineering, leveraging its extensive EPC (Engineering, Procurement, and Construction) experience and modular delivery capabilities in the energy engineering sector, has collaborated with globally leading electrolyser manufacturer Sungrow Hydrogen to jointly develop the MegaFlex solution. Specifically designed for large-scale green hydrogen production, this "factory-as-product" one-stop solution features a single-system capacity of up to 500 MW, with potential for further scaling. Furthermore, it exhibits exceptional flexibility across a wide range of scales, spanning from dozens of megawatts to gigawatt-level green hydrogen applications.

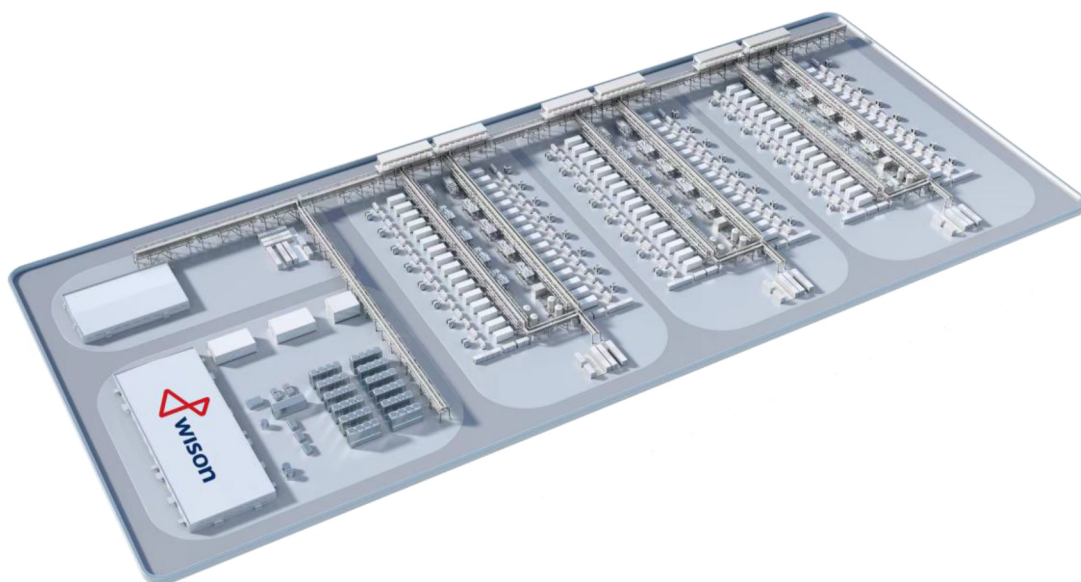


Figure 4.3—MegaFlex Rendering

Wison Engineering's proficiency in modular planning, research, and design enables highly modularized delivery across the entire plant, transforming green hydrogen production facilities from conventional project-based deployments into industrial products that are replicable, scalable, and rapidly deployable. This approach not only facilitates tighter integration between green hydrogen production and CO₂ hydrogenation-to-methanol processes, while simultaneously providing an efficient engineering solution to support low-carbon transformation across traditional industrial sectors.

Conclusion

The 10,000-tonne-per-year demonstration project at Waigaoqiao No.3 Thermal Power Plant in Shanghai has successfully validated the technical feasibility and scalability of CO₂-to-methanol conversion through an integrated system comprising an advanced AEEA-based absorbent, a high-performance modified copper catalyst, an innovative dual-reactor configuration, and an optimised energy cascade. The system achieved an average CO₂ capture rate of 95.1%, methanol selectivity exceeding 99%, and total H₂ and CO₂ conversion rates above 99%, thereby demonstrating not only exceptional technical performance but also establishing a robust technological foundation for future industrial-scale deployment.

With the progressive implementation of China's dual-carbon strategy, green hydrogen-coupled CO₂ capture for methanol production is poised to become a pivotal lever for decarbonizing high-emission industries. While current economic constraints persist due to elevated green hydrogen costs, projected reductions in renewable power generation expenses and electrolyzer technological advancements, coupled with evolving carbon pricing mechanisms (trading, taxation, and credit systems), are expected to enhance green methanol's competitiveness. The Waigaoqiao Third Power Plant demo project exemplifies both technological innovation and China's strategic multi-stakeholder approach to harmonizing energy security with climate objectives. This demonstration provides a scalable industrial paradigm for deep decarbonization for multiple hard-to-abate industries, heralding an emerging circular economy era centered on carbon utilization.

References

- Annual-Report-Working-Group-on-China's-Synergistic-Pathways-for-Carbon-Neutrality-and-Clean-Air. (2022). China's Carbon Neutrality and Clean Air Synergistic Path 2022 - Synergistic Pollution Reduction and Carbon Emission Reduction for Greater Efficiency. Retrieved from

- Aso, D., Orimoto, Y., Higashino, M., Taniguchi, I., & Aoki, Y. (2022). Why does 2-(2-aminoethylamino)ethanol have superior CO₂ separation performance to monoethanolamine? A computational study. *Physical Chemistry Chemical Physics*, **24**(23), 14172–14176.
- Cao, C., Wen, B., Fang, Y., Hou, Z., Wang, Q., Zhang, T., et al (2025). A comprehensive comparison of green ammonia and green methanol from a full chain: production, transportation, storage and utilization. *Carbon Neutral Systems*, **1**(1), 8.
- Chai, S. Y. W., Ngu, L. H., & How, B. S. (2022). Review of carbon capture absorbents for CO₂ utilization. **12**(3), 394–427.
- Du, J., Yang, W., Xu, L., Bei, L., Lei, S., Li, W., et al (2024). Review on post-combustion CO₂ capture by amine blended solvents and aqueous ammonia. *Chemical Engineering Journal*, **488**, 150954.
- EU, (2018). Directive (EU) 2018/2001 on the promotion of the use of energy from renewable sources (recast).
- Guil-López, R., Mota, N., Llorente, J., Millán, E., Pawelec, B., Fierro, J. L. G., et al (2019). Methanol Synthesis from CO₂: A Review of the Latest Developments in Heterogeneous Catalysis. **12**(23), 3902.
- Guoqing, Y., Jucai, Y., Yanli, L., Mingqi, L., Uudam, H., Haiyan, W., et al (2025). Carbon emission reduction accounting and competitiveness analysis of green hydrogen-coupled coal-to-methanol process. *Low-Carbon Chemistry and Chemical Engineering*, **50**, 107–112.
- Huang, J., Jia, S., Chen, W., Wang, Q., Jiang, Y., & Cui, P. (2024). n-Butanol-Regulated Phase Separation of Aminoethylethanolamine (AEEA) as an Efficient Absorbent for CO₂ Capture. *Industrial & Engineering Chemistry Research*, **63**(47), 20688–20696.
- Khan, A. A., Halder, G. N., & Saha, A. K. (2015). Carbon dioxide capture characteristics from flue gas using aqueous 2-amino-2-methyl-1-propanol (AMP) and monoethanolamine (MEA) solutions in packed bed absorption and regeneration columns. *International Journal of Greenhouse Gas Control*, **32**, 15–23.
- Krótki, A., Więclaw-Solny, L., Tatarczuk, A., Spietz, T., Chwoła, T., & Dobras, S. (2023). A pilot study comparing MEA and AEEA solvents in carbon capture. *International Journal of Greenhouse Gas Control*, **126**, 103891.
- Liang, Z., Rongwong, W., Liu, H., Fu, K., Gao, H., Cao, F., et al (2015). Recent progress and new developments in post-combustion carbon-capture technology with amine based solvents. *International Journal of Greenhouse Gas Control*, **40**, 26–54.
- Liu, F., Rochelle, G. T., Wang, T., Chen, E., & Fang, M. (2021). CO₂ absorption rate in biphasic solvent of aminoethylethanolamine and diethylethanolamine. *Chemical Engineering Journal*, **404**, 126503.
- Liu, X., Wang, H., & Lu, J. (2024). Recent progress in understanding the nature of active sites for methanol synthesis over Cu/ZnO catalysts. *Journal of Catalysis*, **436**, 115561.
- Loachamin, D., Casieria, J., Calva, V., Palma-Cando, A., Ávila, E. E., & Ricaurte, M. (2024). Amine-Based Solvents and Additives to Improve the CO₂ Capture Processes: A Review. **8**(6), 129.
- National Development and Reform Commission (NDRC), N. E. A. N. (2022). 14th Five-Year Plan for Modern Energy System. Retrieved from
- NenPower. (2025). Geely's Methanol Vehicles: A New Frontier in Green Energy Emerges. Retrieved from <https://nenpower.com/blog/geelys-methanol-vehicles-a-new-frontier-in-green-energy-emerges/>
- Ochedi, F. O., Yu, J., Yu, H., Liu, Y., & Hussain, A. (2021). Carbon dioxide capture using liquid absorption methods: a review. *Environmental Chemistry Letters*, **19**(1), 77–109.
- Ouda, M., Hank, C., Nestler, F., Hadrich, M., Full, J., Schaadt, A., et al (2019). Power-to-Methanol: Techno-Economical and Ecological Insights. In W. Maus (Ed.), *Zukünftige Kraftstoffe: Energiewende des Transports als ein weltweites Klimaziel* (pp. 380–409). Berlin, Heidelberg: Springer Berlin Heidelberg.
- Rochelle, G. T. (2009). Amine Scrubbing for CO₂ Capture. **325**(5948), 1652–1654.
- Roy, S., Manohar, E. M., Bandyopadhyay, S., Chandra Singh, M., Cha, Y., Giri, S., et al (2025). Advanced catalytic strategies for CO₂ to methanol conversion: noble metal-based heterogeneous and electrochemical approaches. *RSC Sustainability*, **3**(3), 1303–1332.
- Saedy, S., Newton, M. A., Zabilskiy, M., Lee, J. H., Krumeich, F., Ranocchiari, M., et al (2022). Copper–zinc oxide interface as a methanol-selective structure in Cu–ZnO catalyst during catalytic hydrogenation of carbon dioxide to methanol. *Catalysis Science & Technology*, **12**(8), 2703–2716.
- Sholeh Ma'mun, H. S., Olav Julissen. (2007). Absorption characteristics of solvents for carbon dioxide capture. *Jurnal Rekayasa Proses*, **1**(1).
- Stergioudi, F., Baxevani, A., Florou, C., Michailidis, N., Nessi, E., Papadopoulos, A. I., et al (2022). Corrosion Behavior of Stainless Steels in CO₂ Absorption Process Using Aqueous Solution of Monoethanolamine (MEA). *Corrosion and Materials Degradation*, **3**(3), 422–438. doi:10.3390/cmd3030025
- Wang, H., Liu, Y., Laaksonen, A., Krook-Riekkola, A., Yang, Z., Lu, X., et al (2020). Carbon recycling – An immense resource and key to a smart climate engineering: A survey of technologies, cost and impurity impact. *Renewable and Sustainable Energy Reviews*, **131**, 110010.
- Wang, M., Joel, A. S., Ramshaw, C., Eimer, D., & Musa, N. M. (2015). Process intensification for post-combustion CO₂ capture with chemical absorption: A critical review. *Applied Energy*, **158**, 275–291.

- Wang, X., Liu, Y., Wang, Z., Chen, C., Song, Z., Xu, Y., et al (2025). Enhancement of Cu–Zn Interaction via Improved Cu–Zn Contact: Promoting Zinc Migration for Active Site Formation. *Industrial & Engineering Chemistry Research*.
- Wang, X., & Song, C. (2020). Carbon Capture From Flue Gas and the Atmosphere: A Perspective. Volume **8** - 2020.
- Zeng, H., Qu, X., Xu, D., & Luo, Y. (2022). Porous Adsorption Materials for Carbon Dioxide Capture in Industrial Flue Gas. Volume **10** - 2022.
- Zhang, D., Wang, R., Zhang, Z., Yan, H., Zhou, X., Zhao, H., et al (2025). Industrial ultra-low-carbon methanol synthesis routes: techno-economic analysis, life cycle environment assessment and multi-dimensional sustainability evaluation. *Green Chemistry*, **27**(6), 1747–1762.
- Zhenyu Zhao, L. H. (2013). How China Respond to the Carbon Emissions—Challenges, Commitment and Action. *Journal of Low Carbon Economy*, **2**(1), 30–37.
- Zhou, X., Wang, D., Liu, C., Jing, G., Lv, B., & Wang, D. (2024). Enhancing CO(2) capture of an aminoethylethanolamine-based non-aqueous absorbent by using tertiary amine as a proton-transfer mediator: From performance to mechanism. *J Environ Sci (China)*, **140**, 146–156.